head
head

1. **Answer the question below by choosing the correct response.**

   Which of the following expresses $\dfrac{5}{16}$ as a percent?

   A. 0.03125%

   B. 0.3125%

   C. 3.125%

   D. 31.25%

2. Answer the question below by choosing the correct response.

Which of the following is equal to $-2^2 + 2(4-1)^2$

A. 14

B. 22

C. 32

D. 45

3. Answer the question below by choosing the correct response.

A two-dimensional net of a certain three-dimensional figure includes five faces, eight edges, and five vertices. Which of the following three dimensional figures is represented by the net?

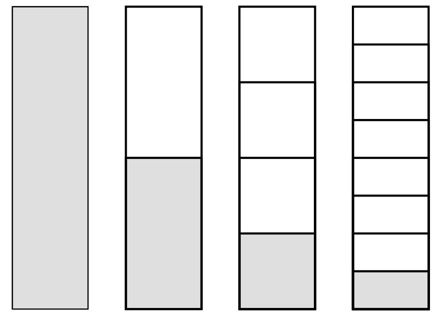
A. Triangular pyramid

B. Triangular prism

C. Rectangular pyramid

D. Rectangular prism

4. Answer the question below by choosing the correct response.



$\frac{1}{1}$ $\frac{1}{2}$ $\frac{1}{4}$ $\frac{1}{8}$

Which of the following is demonstrated by the figure above?

A. When the numerator stays the same and the denominator increases, the fraction increases.

B. When the numerator stays the same and the denominator increases, the fraction decreases.

C. When the denominator stays the same and the numerator increases, the fraction increases.

D. When the denominator stays the same and the numerator increases, the fraction decreases.

5. Answer the question below by choosing the correct response.

Week	Customers
1	51
2	65
3	43
4	115
5	87
6	75
7	120
8	95
9	89

A large department store tallies the number of customer complaints each week for nine consecutive weeks. The number of customer complaints for each week is shown in the provided table. If the number of customers who lodged complaints at the store during the tenth week were included in the data, the mean and median of the data would change but the range would not change. Which of the following could be the number of customers complaints during the tenth week?

A. 35

B. 42

C. 69

D. 129

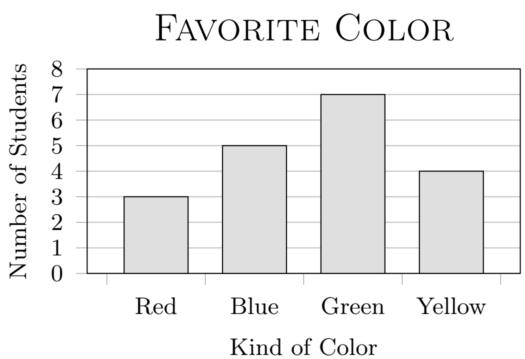
6. Answer the question below by choosing the correct response.

Red	Red	Yellow	Red	Blue
Blue	Green	Red	Blue	Blue
Blue	Green	Yellow	Green	Yellow
Green	Green	Yellow	Green	Green

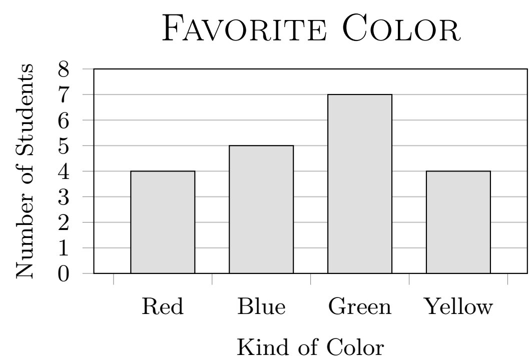
A Pollster asked 20 high-school students to name their favorite color. The students' responses are shown in the table.

The teacher then asked the students to organize and display the data in a bar graph. Which bar graph correctly displays the data?

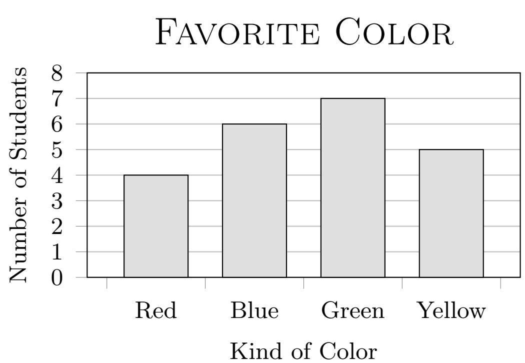
A.



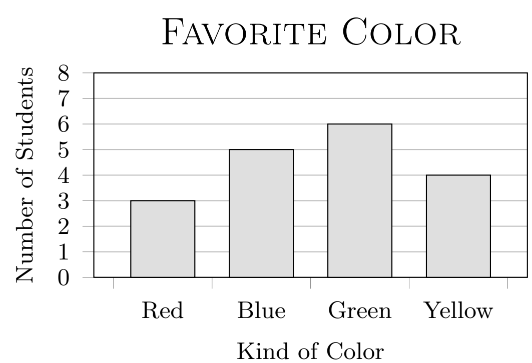
C.



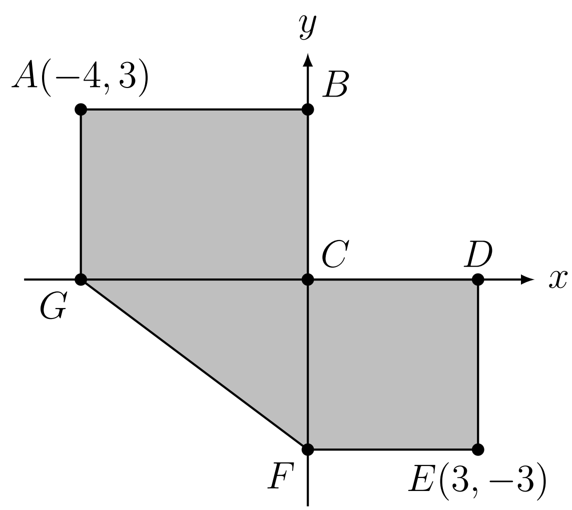
B.



D.



7. Indicate all of your answers.



In the figure shown, the quadrilateral $ABCG$ is a rectangle and the quadrilateral $CDEF$ is a square. Which of the following statements must be true about the figure?

Select **all** that apply.

A. F lies on the y -axis.

B. B lies on the x-axis.

C. A is in the first quadrant.

D. E is in the fourth quadrant

E. The area of the polygon $ABCDEFG$ is 27 square units.

F. The perimeter of the polygon $ABCDEFG$ is 23 units.

8. Answer the question below by choosing the correct response.

Which of the following is equal to $2(100^0)$?

A. 0

B. 1

C. 2

D. 200

9. Answer the question below by choosing the correct response.

A pencil is 21 centimeters in length. How long is the pencil in millimeters?

A. 0.21

B. 2.1

C. 210

D. $2,100$

10. Answer the question below by choosing the correct response.

If the geometric sequence below continues to increase in the same way, what is the next number in the sequence?

$$3, 9, 27, 81, 243 \ldots$$

A. 324

B. 486

C. 729

D. 972

11. Answer the question below by choosing the correct response.

Which of the following lists the coefficients and degrees of the terms in the polynomial shown?

$$4x^6 + 7x^4 - 6$$

A. Coefficients: $4, 7, -6$; Degrees: $6, 4, 0$

B. Coefficients: $4, 7, -6$; Degrees: $6, 4, 1$

C. Coefficients: $2, 4, 0$; Degrees: $4, 7, -6$

D. Coefficients: $2, 4, 1$; Degrees: $4, 7, -6$

12. Answer the question below by choosing the correct response.

In which quadrant is the point $(5, -2)$ located?

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV

13. Answer the question below by choosing the correct response.

Edwins's number: $15,293,863$

Tess's number: $29,018,335$

Both Edwin and Tess wrote a number, as shown above. The digit 9 in Tess's number represents how many times what digit 9 represents in Edwin's number?

A. 100

B. 1,000

C. 10,000

D. 1,000,000

14. Answer the question below by choosing the correct response.

$$4z + 2z$$

What is the value of the algebraic expression shown when $z = 3$

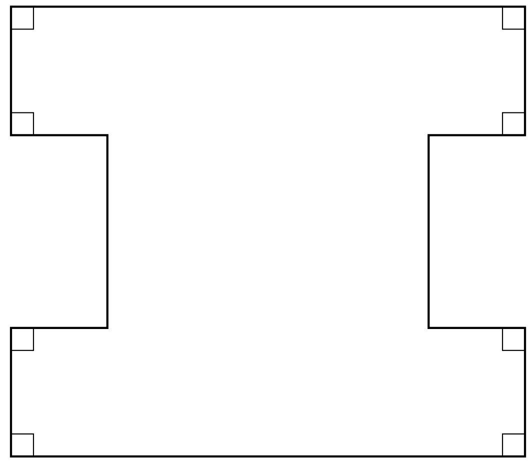
A. 12

B. 18

C. 21

D. 87

15. Answer the question below by choosing the correct response.



Which of the following could be used to describe the polygon shown

A. Regular and convex

B. Regular and concave

C. Irregular and convex

D. Irregular and concave

16. Answer the question below by choosing the correct response.

> Stephen went to the grocery store to buy some food. He bought eight vegetables, half as many fruits as vegetables, and a third as many cans of soup as vegetables. How many items did Stephen buy?

Which of the following statements about the solution to the word problem shown must be true?

A. Because of the real-world context, the solution must belong to the set of all rational positive numbers; therefore the solution is acceptable.

B. Because of the real-world context, the solution must belong to the set of all rational positive numbers; therefore the solution is not acceptable.

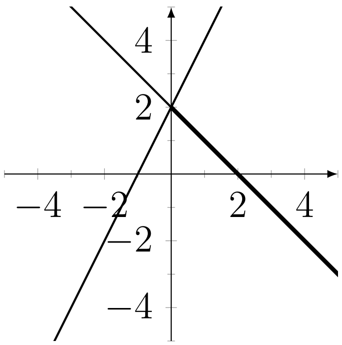
C. Because of the real-world context, the solution must belong to the set of all natural numbers; therefore the solution is acceptable.

D. Because of the real-world context, the solution must belong to the set of all natural numbers; therefore the solution is not acceptable.

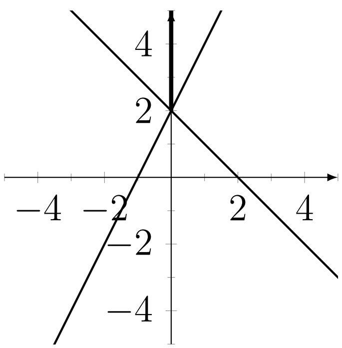
17. Answer the question below by choosing the correct response.

Which of the following graphs in the xy-plane could be used to solve graphically the inequality $-x + 2 > 2(x + 1)$ and shows the solution to the inequality?

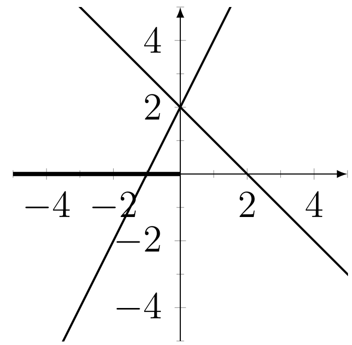
A.



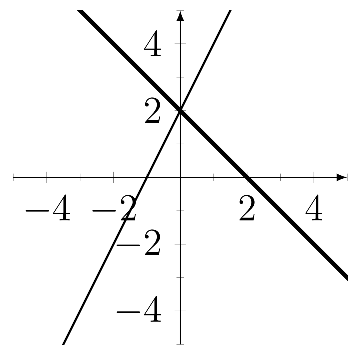
B.



C.



D.



18. Answer the question below by choosing the correct response.

Mrs. Smith gave her students the following table and asked them to determine function rules for the provided data.

x	y
-1	1
0	0

Adeline says the function rule is $y = -x^3$, Brent says that the function rule is $y = -x$, and Chad says that the function rule is $y = -x^2 - 2x$. Which of the three equations could be the function rule for the table?

A. only Brent's

B. only Brent's and Adeline's

C. only Brent's and Chad's

D. Brent's, Adeline's, and Chad's

19. Answer the question below by choosing the correct response.

A public fountain holds of 42,300 liters of water. A crack is leaking at a rate of 75 liters per hour. The equation $\ell = 42,300 - 75h$ can be used to find the number of liters ℓ remaining in the fountain after h days. Which of the following statements is true?

A. ℓ is the independent variable because it is what needs to be found.

B. ℓ is the dependent variable because the volume is dependent on the number of hours h.

C. h is the dependent variable because it is being multiplied by the independent rate of -75.

D. Dependent and independent varaiables cannot be determined in this situation because the equation is linear.

20. Write your answer in the provided form.

Tiana needs to pack 1,120 candies into boxes. If a box can hold 350 candies, how many boxes does she need to pack all the candies.

21. Answer the question below by choosing the correct response.

In the number $756,814$ what is the value of the digit 5?

A. Fifty

B. Five thousand

C. Ten thousand

D. Fifty thousand

22. Answer the question below by choosing the correct response.

Kathryn setup a saving's account by making an initial deposit of \$120. Each month, Kathryn deposits \$24 into the savings account. The amount of money she has saved after x weeks can be represented with the expression $24x + 120$. This expression is equivalent to which of the following expressions?

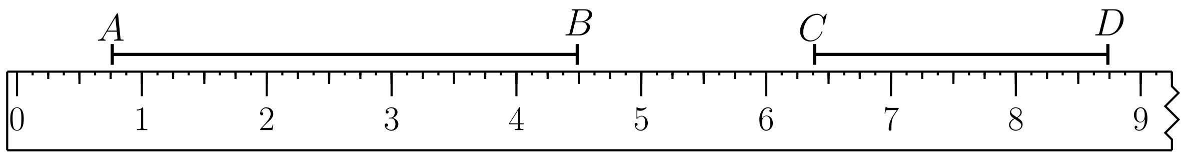
A. $4(6x + 120)$

B. $4(6x + 20)$

C. $6(4x + 20)$

D. $6(4x + 100)$

23. Answer the question below by choosing the correct response.



Note: Figure not drawn to scale

The figure shows a ruler with inches marked and two line segments drawn. To the nearest eight of an inch, how much longer, in inches, is the length of $\overline{AB}$ than the length of $\overline{CD}$?

A. $1\frac{3}{8}$

B. $1\frac{1}{2}$

C. $1\frac{5}{8}$

D. 2

E. 3.75

24. Answer the question below by choosing the correct response.

Which of the following numbers is the greatest?

A. 0.1031

B. 0.1004

C. 0.105

D. 0.10009

25. Answer the question below by choosing the correct response.

In February, 20 percent of the 3750 students at a high schools had a perfect attendance record. Which of the following computations can be used to determine the number of students with a perfect attendance record?

A. $20 \times 3{,}750$

B. $\dfrac{20}{3{,}750}$

C. $\dfrac{1}{20} \times 3{,}750$

D. $\dfrac{1}{5} \times 3{,}750$

26. Answer the question below by choosing the correct response.

In the formula, $d = r \times t$, if d equals 75 and t remains constant, which of the following is equivalent to r?

A. $75t$

B. $\frac{75}{t}$

C. $\frac{t}{75}$

D. $\frac{d}{75t}$

27. Answer the question below by choosing the correct response.

$$17(4-7) = 17 \times 4 - 17 \times 7$$

The equation shown demonstrates which of the following?

A. The distributive property of multiplication over addition.

B. The commutative property of multiplication.

C. The associative property of multiplication.

D. The additive inverse and additive identity properties.

28. Answer the question below by choosing the correct response.

Which of the following is the product of two odd numbers and an even number, each of which is greater than 1?

A. 16

B. 24

C. 27

D. 30

29. Answer the question below by choosing the correct response.

A pet food company makes 12-kilogram and 30-kilogram bags of dog food. For the 12-kilogram bag, the company uses 5-kilograms of chicken meat, and the rest is corn. If the proportion of chicken meat is the same in the 30-kilogram bag as the 12-kilogram bag, how many kilograms of corn does the 30-kilogram bag have?

A. 2.8

B. 12.5

C. 17.5

D. 20

30. Answer the question below by choosing the correct response.

Step 1: Select a numerical value for the variable b.
Step 2: Add 5 to b.
Step 3: Multiply the result by 2.
Step 4: Square the result.
Step 5: End

Consider the algorithm shown above. Following the algorithm is equivalent to evaluating which of the following algebraic expressions?

A. $(2b + 5)^2$

B. $2(b + 5)^2$

C. $2^2(b + 5)$

D. $2^2(b + 5)^2$

31. Answer the question below by choosing the correct response.

The first two terms of the sequence shown are 3 and 4. Each subsequent term, beginning with the third, is found by adding the two preceding terms and multiplying the sum by -2. What is n_5, the fifth term of the sequence?

$$3, 4, n_3, n_4, n_5, \ldots$$

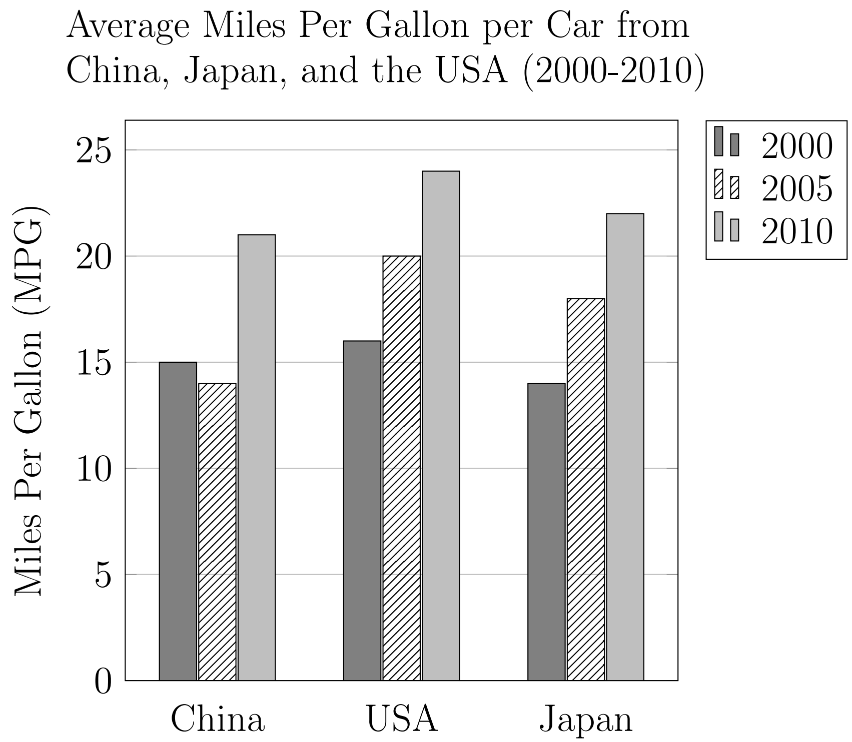
A. -16

B. -14

C. -12

D. -10

32. Indicate all of your answers.



Which of the following statements can be inferred from the graph shown?

A. For each country, the average MPG increased each year from the previous year

B. The country that had the greatest yearly average MPG for each of the three years shown had a three year total between 55 and 65 MPG.

C. The average MPG in China increase by more than one third from 2000 to 2010.

D. The average MPG in the USA in 2010 was more than double of that in China in 2000.

33. Answer the question below by choosing the correct response.

x	$f(x)$
60	
25	-5
20	-2
15	1

Values of the linear function f are provided in the table shown. Find $f(60)$.

A. -8

B. -14

C. -20

D. -26

34. Answer the question below by choosing the correct response.

Wei tossed a fair coin 12 times with an outcome of $H, T, H, T, T, T, H, H, T, T, T, T$, where H means the coin landed on heads and T means the coin landed on H. Approximately what is the probability that the next toss will be T?

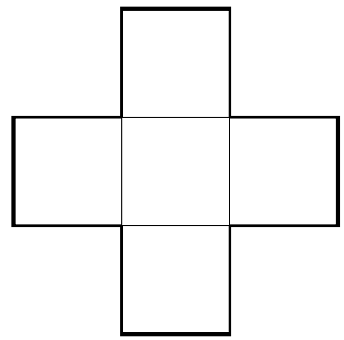
A. 0.67

B. 0.5

C. 0.33

D. 0.83

35. Answer the question below by choosing the correct response.



If the perimeter of each of the five regular quadrilaterals in the figure shown is 24, what is the perimeter of the figure?

A. 72

B. 84

C. 120

D. 144

36. Answer the question below by choosing the correct response.

The population of a certain county was 200,000 people. Ten years later the population of the same county had decreased to 192,000. Approximately, what was the percent decrease in the county's population in that ten year period?

A. 4%

B. 4.17%

C. 40%

D. 41.7%

37. Answer the question below by choosing the correct response.

Which of the following is an example of the associative property of addition?

A. $(a+b)+c=(b+a)+c$

B. $ab+c=ba+c$

C. $(a+b)+c=a+(b+c)$

D. $a(b+c)=ab+ac$

38. Answer the question below by choosing the correct response.

What is the greatest odd factor of 1,344?

A. 3

B. 7

C. 21

D. 113

39. Answer the question below by choosing the correct response.

At a school bake sale, Ted sold pies for \$5.00 each and cookies for \$1.50 each. Nancy spent a total of \$16.50 on Ted's pies and cookies. If Nancy bought at least one cookie and one pie, how many pies did she buy?

A. 1

B. 2

C. 3

D. 4

40. Answer the question below by choosing the correct response.

The Carolina Panthers football team scored an average of 35 points in 5 games. In the first four games, the team scored 28, 34, 35, and 40 points. How many points did they score in their fifth game?

A. 30

B. 36

C. 37

D. 38

41. Answer the question below by choosing the correct response.

On Gwen's map, 30 miles is represented by 1 inch, and on Luke's map, 40 miles is represented by 1 inch. The area represented by a 1-inch by 1-inch square is how many more square miles on Luke's map than on Gwen's map?

A. 10

B. 700

C. 1600

D. 2500

42. Indicate all of your answers.

Which of the following are equivalent to dividing 306 by 18

Select **all** that apply.

A. $(306 \div 2) \div 9$

B. $2(153 \div 18)$

C. $(153 \div 9) + (153 \div 9)$

D. $(260 \div 18) + (46 \div 18)$

43. Answer the question below by choosing the correct response.

Item	Cost
Hammer	\$5.53
Nails	\$5.09
Bucket	\$5.99
Brush	\$5.09
Pencil	\$1.79
Washers	\$5.29

Prices include sales tax

Joe goes to the hardware store and purchases one of each item in the list above at the indicated prices. If Joe pays with two \$20 bills, which of the following is closest to the change he would receive?

A. \$11

B. \$12

C. \$13

D. \$14

44. Answer the question below by choosing the correct response.

Which of the following has the greatest value?

A. 7,200 ones

B. 708 tens

C. 73 hundreds

D. 7 thousands

45. Answer the question below by choosing the correct response.

A flag pole casts a shadow that is 23 feet long. Nearby, a stop sign with a height of 7 feet casts a shadow that is 3 feet long. Assuming that flag pole and stop sign are both perpendicular to the ground, the height of the flag pole must be between which of the following values?

A. 0 and 10 feet

B. 40 and 50 feet

C. 50 and 60 feet

D. 170 and 180 feet

46. Answer the question below by choosing the correct response.

Bonnie's night work shift began at $7:53$ P.M. and ended at $5:29$ A.M. How long did her shift last?

A. 8 hours and 22

B. 8 hours and 36 minutes

C. 9 hours and 22 minutes

D. 9 hours and 36 minutes

47. Answer the question below by choosing the correct response.

Suppose that it takes 3 hours to mow an acre of grass. Which of the following expression represents the number of acres that can be mowed in h hours.

A. $3h$

B. $\frac{h}{3}$

C. $\frac{h}{60 \times 3}$

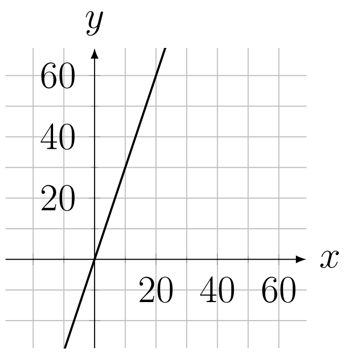
D. $\frac{60 \times h}{3}$

48. Answer the question below by choosing the correct response.

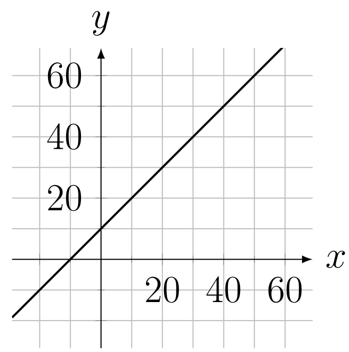
x	y
-20	-10
-10	0
10	20
20	30

Which of the following graphs best represents the relationship between x and y in the table above.

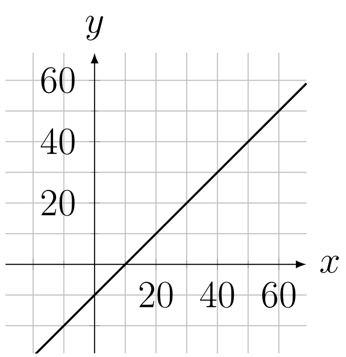
A.



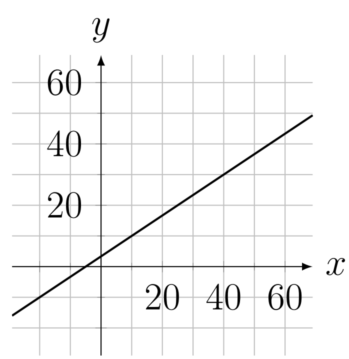
C.



B.



D.



49. Answer the question below by choosing the correct response.

If $75 - 3x = 6y$, find x in terms of y.

A. $x = 3(6y - 75)$

B. $x = 3(6y + 75)$

C. $x = -(6y - 75) \div 3$

D. $x = (6y + 75) \div 3$

50. Answer the question below by choosing the correct response.

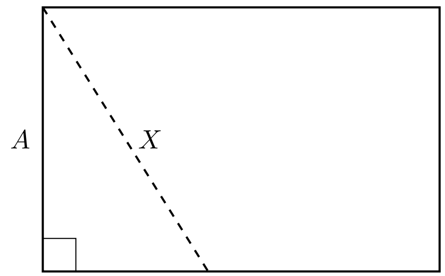


FIGURE 1

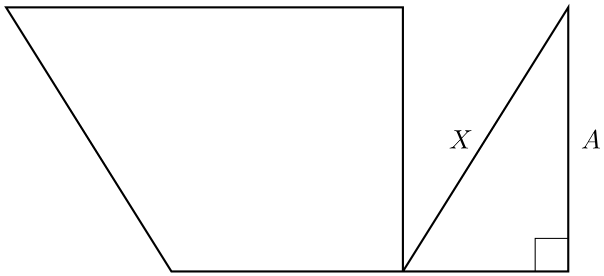


FIGURE 2

The shape shown in Figure 1 is a rectangle. Suppose that the rectangle is cut along the dashed line and the resulting shapes are rearranged as shown in Figure 2. Which of the following statements regarding the resulting shape is true?

A. The area of Figure 1 is less than the area of Figure 2, and the perimeter of Figure 1 is equal to the perimeter of Figure 2.

B. The area of Figure 1 is less than the area of Figure 2, and the perimeter of Figure 1 is less than the perimeter of Figure 2.

C. The area of Figure 1 is equal to the area of Figure 2, and the perimeter of Figure 1 is equal to the perimeter of Figure 2.

D. The area of Figure 1 is equal to the area of Figure 2, and the perimeter of Figure 1 is less than the perimeter of Figure 2.